

WEEK 2 – Data Visibility & Exploration

Goal

👉 Understand data deeply + create basic analytics

INPUT

- Bronze tables:
 - `bronze_claims_raw`
 - `bronze_provider_raw`
 - `bronze_diagnosis_raw`
 - `bronze_cost_raw`
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PROCESS

Step 1: Explore Data

- Check:
 - top rows
 - unique values
 - distribution
-

Step 2: Basic SQL Analysis

- Example:
 - total claims
 - claims per provider
 - avg billed amount
 - claims per diagnosis
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Step 3: Join Data (Basic)

- Join:
 - claims + provider
 - claims + diagnosis
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Step 4: Create Simple Views

- Example:
 - claims by specialty
 - claims by region
 - high-cost claims
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Step 5: Build Dashboard (Basic)

- Use:
 - Databricks / Power BI
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OUTPUT

You must have:

- ✓ Summary tables
- ✓ Basic joins working
- ✓ Dashboard showing:

- total claims
 - cost trends
 - provider activity
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FINAL OUTPUT

Raw Data → Insights (no cleaning yet)



WEEK 3 – Data Cleaning & Silver Layer



Goal

👉 Clean data + create trusted dataset (Silver layer)



INPUT

- Bronze tables
-



PROCESS

Step 1: Handle Missing Values

- Fill / remove:
 - missing diagnosis
 - missing provider
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Step 2: Remove Duplicates

- Ensure:
 - unique claims
 - unique providers
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Step 3: Fix Data Types

- Convert:
 - dates → proper format
 - amounts → numeric
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Step 4: Standardize Data

- Fix:
 - ICD/CPT format
 - text case
 - spacing
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Step 5: Create Silver Tables

Create:

silver_claims
silver_provider
silver_diagnosis
silver_cost

Step 6: Validate Data

- Check:
 - no null critical fields
 - correct joins
 - consistent values
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OUTPUT

You must have:

- ✓ Clean datasets
 - ✓ Silver tables
 - ✓ Data quality improved
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FINAL OUTPUT

Raw Data → Clean Data (Ready for features)



FINAL SUMMARY

Week	Focus	Output
Week 2	Explore data	Insights + dashboard
Week 3	Clean data	Silver tables



ONE LINE



Week 2 = Understand data
Week 3 = Fix data